

Class nr. 18

Subject: 2. Intertemporal choice

2.1. Consumer's intertemporal choice (practice class)

Exercise 10.1. from Bergstrom and Varian's book (2006) "Workouts in Intermediate Microeconomics", p. 130

Peregrine Pickle consumes (c_1, c_2) and earns (m_1, m_2) in periods 1 and 2 respectively. Suppose the interest rate is r .

- a) Write down Peregrine's intertemporal budget constraint in present value terms.
- b) If Peregrine does not consume anything in period 1, what is the maximum value he can consume in period 2, that is, what is the future value of his initial endowment?
- c) If Peregrine does not consume anything in period 2, what is the maximum value he can consume in period 1, that is, what is the present value of his initial endowment? What is the slope of his budget constraint?

Exercise 10.3. from Bergstrom and Varian's book (2006) "Workouts in Intermediate Microeconomics", pp. 131-132

Nickleby has an income of \$2,000 this year, and he expects an income of \$1,100 next year. He can borrow and lend money at an interest rate of 10%. Consumption goods cost \$1 per unit this year and there is no inflation.

- a) What is the present and future value of Nickleby's endowment? Sketch, in a graph, his endowment, budget constraint and budget set.
- b) Suppose that Nickleby has the utility function $U(C_1, C_2) = C_1 C_2$. Write an expression for Nickleby's marginal rate of substitution between consumption this year and consumption next year.
- c) What is the slope of Nickleby's budget constraint? Write an equation for his optimal condition (this equation states that the slope of Nickleby's indifference curve is equal to the slope of his budget constraint when the interest rate is 10%). Also write down Nickleby's budget constraint.

- d) Solve the two previous equations in order to determine Nickleby's optimal intertemporal consumption bundle (combination of present and future consumptions).
- e) Will he borrow or save in the first period? How much?
- f) Sketch in a graph the combinations of present and future consumption that Nickleby will not choose if the interest rate increases to 20%.
- g) Solve for Nickleby's optimal choice when the interest rate is 20%.
- h) In this case, will he borrow or save? How much?

Exercise 10.10. from Bergstrom and Varian's book (2006) "Workouts in Intermediate Microeconomics", pp. 138-139

In an isolated mountain village, the only crop is corn. Good harvests alternate with bad harvests. This year the harvest will be 1,000 bushels¹. Next year it will be 150 bushels. There is no trade with the outside world. Corn can be stored from one year to the next, but rats will eat 25% of what is stored in a year. The villagers have Cobb-Douglas utility functions, $U(c_1, c_2) = c_1 c_2$ where c_1 is consumption this year, and c_2 is consumption next year.

- a) Sketch in a graph the consumption possibilities (budget set) for the village.
- b) How much corn will the villagers consume this year? How much will the rats eat? How much corn will the villagers consume next year?
- c) Suppose that a road is built to the village so that now the village is able to trade with the rest of the world. Now the villagers are able to buy and sell corn at the world price, which is \$1 per bushel. They are also able to borrow and lend money at an interest rate of 10%. Sketch (in a graph) the new budget constraint. Determine the new optimal consumption point for both periods.
- d) Suppose that all is as in question c) except that now there is a transportation cost of \$0.1 per bushel for every bushel of grain hauled into or out of the village. Sketch in a graph the new budget constraint for the village under these new circumstances.

¹ A measure of volume used in the United States that is roughly equivalent to 35 liters.